

35.3 Separation

Proposition 35.1.3 says the logistic log-likelihood is strictly concave when the model matrix has full column rank. That guarantees at most one maximum, not at least one, and for binary data the failure is common: it happens whenever some linear combination of the covariates predicts the response perfectly, as in small samples, with rare outcomes, and whenever a categorical covariate has a level with only one outcome in it.

35.3.1 Three configurations

Write $s_i = 2y_i - 1 \in \{-1, 1\}$ for the signed response of an ungrouped binary observation. The logistic log-likelihood is then

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n \{y_i \log \pi_i + (1 - y_i) \log(1 - \pi_i)\} = - \sum_{i=1}^n \log \{1 + \exp(-s_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta})\}, \quad (35.3.1)$$

because $\pi_i = 1/\{1 + \exp(-\mathbf{x}_{(i)}^\top \boldsymbol{\beta})\}$ and $1 - \pi_i$ is the same expression with the sign of the linear predictor reversed. Every term is negative and rises towards zero as $s_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ increases, so the whole question is whether some direction raises all of them at once.

Definition 35.3.1 — Separation and overlap

Let $\mathbf{X}$ be $n \times p$ with rows $\mathbf{x}_{(i)}^\top$, and let $s_i = 2y_i - 1$. The data are

- (a) **completely separated** if there is a $\mathbf{b} \neq \mathbf{0}$ with $s_i \mathbf{x}_{(i)}^\top \mathbf{b} > 0$ for every i ;
- (b) **quasi-completely separated** if they are not completely separated but there is a $\mathbf{b} \neq \mathbf{0}$ with $s_i \mathbf{x}_{(i)}^\top \mathbf{b} \geq 0$ for every i ;
- (c) in **overlap** if $\mathbf{b} = \mathbf{0}$ is the only $\mathbf{b}$ with $s_i \mathbf{x}_{(i)}^\top \mathbf{b} \geq 0$ at every observation.

A vector $\mathbf{b}$ of the kind in (a) or (b) is a **separating direction**.

The three cases are mutually exclusive and exhaustive, and a linear programme decides which holds: maximize t over $\mathbf{b}$ in the unit cube subject to $s_i \mathbf{x}_{(i)}^\top \mathbf{b} \geq t$. A positive optimum is complete separation; an optimum of zero attained only at $\mathbf{b} = \mathbf{0}$ is overlap, and otherwise quasi-complete separation.

35.3.2 Existence

Theorem 35.3.1 — Existence and uniqueness of the maximum likelihood estimate

Let $\text{rank}(\mathbf{X}) = p$ and let ℓ be the logistic log-likelihood of Eq. 35.3.1. Then:

- (a) if the data are completely or quasi-completely separated, ℓ has no maximizer: for

- every β there is a β' with $\ell(\beta') > \ell(\beta)$;
- (b) under complete separation $\sup_{\beta} \ell(\beta) = 0$, and the supremum is approached along any separating ray $\beta = t\mathbf{b}$, $t \rightarrow \infty$, with every fitted probability tending to 0 or 1;
- (c) if the data are in overlap, ℓ attains its supremum at exactly one $\hat{\beta} \in \mathbb{R}^p$, and $\hat{\beta}$ solves Eq. 35.1.2.

Proof.

- (a) Let $\mathbf{b} \neq \mathbf{0}$ satisfy $s_i \mathbf{x}_{(i)}^\top \mathbf{b} \geq 0$ for all i . Not every one of these numbers can be zero: if $\mathbf{x}_{(i)}^\top \mathbf{b} = 0$ for all i then $\mathbf{X}\mathbf{b} = \mathbf{0}$, and $\text{rank}(\mathbf{X}) = p$ forces $\mathbf{b} = \mathbf{0}$. So $s_k \mathbf{x}_{(k)}^\top \mathbf{b} > 0$ for at least one k . Each summand of Eq. 35.3.1 is a strictly decreasing function of $-s_i \mathbf{x}_{(i)}^\top \beta$, hence a nondecreasing function of t along $\beta + t\mathbf{b}$, and the k th is strictly increasing. Therefore $\ell(\beta + \mathbf{b}) > \ell(\beta)$ for every β , and no maximizer exists.
- (b) Under complete separation put $\delta = \min_i s_i \mathbf{x}_{(i)}^\top \mathbf{b} > 0$. Then $\ell(t\mathbf{b}) \geq -n \log(1 + e^{-t\delta}) \rightarrow 0$, while $\ell < 0$ everywhere, so the supremum is 0 and is not attained. Since $s_i \mathbf{x}_{(i)}^\top(t\mathbf{b}) \rightarrow \infty$, the fitted probability π_i tends to 1 when $y_i = 1$ and to 0 when $y_i = 0$.
- (c) First, ℓ is strictly concave. Its Hessian is $-\mathbf{X}^\top \mathbf{W} \mathbf{X}$ with $\mathbf{W} = \text{diag}\{\pi_i(1 - \pi_i)\}$ positive definite (Proposition 35.1.3), and $\mathbf{b}^\top \mathbf{X}^\top \mathbf{W} \mathbf{X} \mathbf{b} = 0$ forces $\mathbf{X}\mathbf{b} = \mathbf{0}$, hence $\mathbf{b} = \mathbf{0}$. Second, ℓ is coercive: $\ell(\beta) \rightarrow -\infty$ as $\|\beta\| \rightarrow \infty$. Suppose not. Then there are β_k with $\|\beta_k\| \rightarrow \infty$ and $\ell(\beta_k) \geq -c$ for some $c < \infty$. Put $\mathbf{b}_k = \beta_k / \|\beta_k\|$; since the unit sphere is compact, a subsequence converges to some $\mathbf{b}$ with $\|\mathbf{b}\| = 1$. If $s_i \mathbf{x}_{(i)}^\top \mathbf{b} < 0$ for some i , then along that subsequence $s_i \mathbf{x}_{(i)}^\top \beta_k = \|\beta_k\| s_i \mathbf{x}_{(i)}^\top \mathbf{b}_k \rightarrow -\infty$, so the i th term of Eq. 35.3.1 tends to $-\infty$; as all the other terms are at most 0, this gives $\ell(\beta_k) \rightarrow -\infty$, a contradiction. Hence $s_i \mathbf{x}_{(i)}^\top \mathbf{b} \geq 0$ for every i with $\mathbf{b} \neq \mathbf{0}$, contradicting overlap. So ℓ is coercive, and the set $\{\beta : \ell(\beta) \geq \ell(\mathbf{0})\}$ is closed and bounded. A continuous function attains its maximum on a compact set, and strict concavity makes the maximizer unique. It is an interior maximum of a differentiable function, so the gradient vanishes there, which is Eq. 35.1.2. ■

Parts (a) and (c) say that **the maximum likelihood estimate exists if and only if the data are in overlap**, a characterization due to Silvapulle (1981) and, in this form, to Albert and Anderson (1984).

Remark (What quasi-complete separation leaves behind). Under quasi-complete separation let $\mathbf{b}$ be a separating direction and $S = \{i : \mathbf{x}_{(i)}^\top \mathbf{b} = 0\}$. Along $\beta + t\mathbf{b}$ the terms outside S tend to zero and those inside do not move, so the supremum of ℓ is that of the subsample S alone, approached with the coefficients along $\mathbf{b}$ running off to infinity while the rest settle down: which is why software reports some coefficients that look sensible alongside others in the tens with standard errors in the millions.

35.3.3 What the computer does

Fisher scoring does not know any of this: given separated data it climbs the ray until it hits the iteration limit or the weights underflow, so the size of the reported coefficients is set by the iteration count, and the standard errors are larger still, because $(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$ explodes as the

weights vanish.

Example. (A separated subgroup)

Of the 944 respondents of Example (Voting intention in the 1996 American election), 13 reported that their schooling stopped at or before the eighth grade. Among them party identification predicts the intended vote perfectly: the ten at 0, 1 or 2 on the party scale all expected to vote Clinton, the three at 4 or 6 all Dole. The direction $\mathbf{b} = (-3, 1)^\top$, the linear predictor party -3 , satisfies $s_i \mathbf{x}_{(i)}^\top \mathbf{b} > 0$ for all thirteen, so the data are completely separated and no maximum likelihood estimate exists.

Asked to fit logit $\pi = \beta_0 + \beta_1 \text{ party}$, software stopped at ten iterations reports $\hat{\beta}_1 = 8.75$ with standard error 29.3; at twenty-five iterations it reports 23.75 with a standard error of order 10^4 . The deviance falls below 10^{-3} after ten steps and below 10^{-9} after twenty-five, and the Wald test is nowhere near significant at either stopping point.

Inference is nevertheless possible, from the likelihood ratio. The profile log-likelihood rises to its supremum of zero, so the set of β_1 not rejected at the five per cent level is the half-line $(0.902, \infty)$: a one-sided interval, correctly reporting that the data bound the effect from below and not from above.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm

anes = sm.datasets.anes96.load_pandas().data
sub = anes[anes["educ"] == 1]                # schooling stopped at or before
      grade eight
X = sm.add_constant(sub[["PID"]]).to_numpy()
y = sub["vote"].to_numpy()

print(pd.crosstab(sub["PID"], sub["vote"]))
for maxiter in (5, 10, 25, 50):
    f = sm.GLM(y, X, family=sm.families.Binomial()).fit(maxiter=maxiter, tol=1e-16)
    print(f"{maxiter:3d} steps: slope {f.params[1]:10.3f}    standard error {f.bse
          [1]:14.3f}"
          f"    deviance {f.deviance:.2e}")
```

[Run this code online](#)

Example. (An empty cell)

Among the 127 respondents with a doctorate, model the vote by party identification as a seven-level factor with the independent-Republican category (PID = 4) as baseline, that being the only middle level with both outcomes present. Three levels have no variation at all, and exactly 3 of the six contrasts run off to infinity. The others behave normally: the contrast for weak Democrats is -3.140 with standard error 0.916. The deviance, 53.808, is well defined, because the diverging coefficients drive their own cells' fitted probabilities to 0 or 1, contributing nothing to it.

35.3.4 Penalized likelihood

The estimate fails to exist because the likelihood has no interior maximum, so every remedy adds something that does; three are in general use. Firth (1993) proposed adding half the log determinant of the Fisher information to the log-likelihood, the log of the Jeffreys prior density, as a way of removing the leading $O(n^{-1})$ bias of the maximum likelihood estimate; that it also cures separation was noticed later, by Heinze and Schemper (2002). The modified score equations take a memorable form.

Proposition 35.3.2 — Firth's modified score

For ungrouped logistic regression with $\text{rank}(\mathbf{X}) = p$, let $\ell^*(\boldsymbol{\beta}) = \ell(\boldsymbol{\beta}) + \frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{W} \mathbf{X})$ with $\mathbf{W} = \text{diag}\{\pi_i(1 - \pi_i)\}$. Let $h_i = \pi_i(1 - \pi_i) \mathbf{x}_{(i)}^\top (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{x}_{(i)}$ be the i th diagonal entry of the weighted hat matrix $\mathbf{W}^{1/2} \mathbf{X} (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}^{1/2}$. Then

$$\frac{\partial \ell^*}{\partial \boldsymbol{\beta}} = \mathbf{X}^\top \left\{ \mathbf{y} - \boldsymbol{\pi} + \mathbf{h} \circ \left(\frac{1}{2} \mathbf{1} - \boldsymbol{\pi} \right) \right\}, \quad (35.3.2)$$

where $\circ$ is the entrywise product. Equivalently, the Firth estimate is a fixed point: it is the ordinary maximum likelihood estimate for pseudo-data in which observation i has $y_i + h_i/2$ successes out of $1 + h_i$ trials, with h_i evaluated at that same estimate. Along any ray $t\mathbf{b}$ that completely separates the data, $\ell^*(t\mathbf{b}) \rightarrow -\infty$.

Proof. Write $F(\boldsymbol{\beta}) = \mathbf{X}^\top \mathbf{W} \mathbf{X}$ and $w_i = \pi_i(1 - \pi_i)$. Since $\partial \pi_i / \partial \beta_j = x_{ij} w_i$ and $dw/d\pi = 1 - 2\pi$,

$$\frac{\partial w_i}{\partial \beta_j} = x_{ij} w_i (1 - 2\pi_i), \quad \frac{\partial F}{\partial \beta_j} = \mathbf{X}^\top \mathbf{D}_j \mathbf{X},$$

with $\mathbf{D}_j = \text{diag}\{x_{ij} w_i (1 - 2\pi_i)\}$. By the derivative of a log determinant,

$$\begin{aligned} \frac{\partial \log \det F}{\partial \beta_j} &= \text{tr}(F^{-1} \mathbf{X}^\top \mathbf{D}_j \mathbf{X}) = \sum_i x_{ij} w_i (1 - 2\pi_i) \mathbf{x}_{(i)}^\top F^{-1} \mathbf{x}_{(i)} \\ &= \sum_i x_{ij} (1 - 2\pi_i) h_i. \end{aligned}$$

Halving and adding the ordinary score $\sum_i x_{ij} (y_i - \pi_i)$ gives Eq. 35.3.2, since $\frac{1}{2}(1 - 2\pi_i) = \frac{1}{2} - \pi_i$. For the pseudo-data statement, the ordinary score for $y_i^* = y_i + h_i/2$ successes out of $m_i^* = 1 + h_i$ trials is $\sum_i x_{ij} (y_i^* - m_i^* \pi_i)$, which is the same expression.

For the last claim, let $\delta = \min_i s_i \mathbf{x}_{(i)}^\top \mathbf{b} > 0$. At $\boldsymbol{\beta} = t\mathbf{b}$ every fitted probability is within $e^{-t\delta}$ of 0 or 1, so $w_i \leq e^{-t\delta}$ and $\mathbf{X}^\top \mathbf{W} \mathbf{X} \preceq e^{-t\delta} \mathbf{X}^\top \mathbf{X}$ in the nonnegative definite order. Hence $\log \det(\mathbf{X}^\top \mathbf{W} \mathbf{X}) \leq -pt\delta + \log \det(\mathbf{X}^\top \mathbf{X})$, while $\ell \leq 0$. Therefore $\ell^*(t\mathbf{b}) \leq -\frac{1}{2}pt\delta + \text{constant} \rightarrow -\infty$. ■

The penalty pulls the estimate towards zero exactly where the likelihood is flat: along the offending ray it falls linearly while the likelihood has nothing left to gain. That the Firth estimate is finite for *every* configuration with $\text{rank}(\mathbf{X}) = p$ was proved by Kosmidis and Firth (2021), by a longer argument.

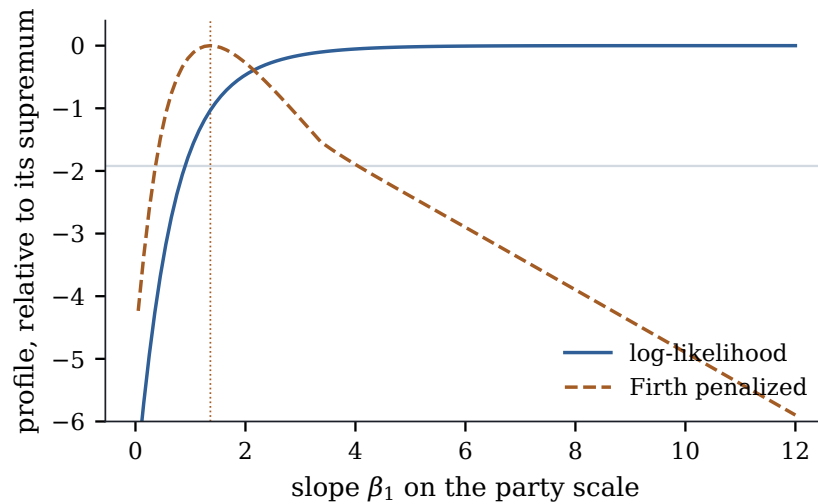


Figure 35.3.1: Profile log-likelihoods for the slope in [Example \(A separated subgroup\)](#), each relative to its own supremum. The ordinary profile (solid) increases to zero and is never attained, so the interval cut out by the horizontal line at $-\chi_{0.95}^2(1)/2$ is a half-line. Firth's penalized profile (dashed) has an interior maximum, marked by the vertical line.

Example. (The separated subgroup, penalized)

Applied to [Example \(A separated subgroup\)](#), Firth's method gives $\hat{\beta}_1 = 1.362$ with standard error 0.708 and intercept -4.049 . The Wald statistic is 1.92, short of the conventional threshold, but the interval from inverting the *penalized* likelihood ratio is (0.364, 4.045), which excludes zero comfortably. With thirteen observations and a log-likelihood as asymmetric as [Figure 35.3.1](#) shows, the quadratic approximation behind the Wald statistic is simply not available.

```
def firth(X, y, steps=200):
    """Maximize the log-likelihood penalized by (1/2) log det(X'WX): Firth's
    estimate."""
    beta = np.zeros(X.shape[1])
    for _ in range(steps):
        mu = 1 / (1 + np.exp(-X @ beta))
        w = mu * (1 - mu)
        F = X.T @ (w[:, None] * X)                # the expected
        information
        h = w * np.einsum("ij,jk,ik->i", X, np.linalg.inv(F), X)
        beta = beta + np.linalg.solve(F, X.T @ (y - mu + h * (0.5 - mu)))
    return beta

beta_firth = firth(X, y)
print("Firth estimate:", beta_firth.round(3))
```

[Run this code online](#)

35.3.5 Ridge and Cauchy priors

A quadratic penalty works too, for a simpler reason. Maximizing $\ell(\boldsymbol{\beta}) - \frac{1}{2}\lambda\|\boldsymbol{\beta}\|^2$ is the logistic analogue of ridge regression (Definition 27.2.1), and the objective is strictly concave and coercive whatever the data, so a unique maximizer always exists (Exercise (B1)). It is the posterior mode under independent $N(0, \lambda^{-1})$ priors, which is how Chapter 30 reads every penalty in Proposition 30.1.4. In practice, and in the code below, the intercept is left out of the penalty, so that the prior reading is a flat one there; the maximizer still exists, provided both outcomes occur (Exercise (B1)). The cost is bias towards zero that depends on the scale of the covariates, and a λ that must come from somewhere, usually cross-validation (Chapter 29).

```
def ridge_logistic(X, y, lam, steps=200):
    """Maximize the log-likelihood minus (lam/2) times the squared norm of the
    slopes."""
    d = np.ones(X.shape[1])
    d[0] = 0.0                                     # leave the intercept
    unpenalized
    beta = np.zeros(X.shape[1])
    for _ in range(steps):
        mu = 1 / (1 + np.exp(-X @ beta))
        w = mu * (1 - mu)
        H = X.T @ (w[:, None] * X) + lam * np.diag(d)
        beta = beta + np.linalg.solve(H, X.T @ (y - mu) - lam * d * beta)
    return beta

for lam in (0.1, 1.0, 10.0):
    print(f"ridge, lambda = {lam:4.1f}: {ridge_logistic(X, y, lam).round(3)}")
```

[Run this code online](#)

On the separated subgroup the ridge estimate of the slope is 2.601 at $\lambda = 0.1$, 1.356 at $\lambda = 1$ and 0.502 at $\lambda = 10$: the whole range is “the answer” for some penalty, which is the point. Gelman, Jakulin, Pittau and Su (2008) proposed instead independent Cauchy priors of scale 2.5 on standardized covariates, whose heavy tails leave a genuinely large coefficient nearly alone. Firth’s penalty alone is invariant to how the covariates are parameterized.

35.3.6 Exact conditional inference

A third route abandons the normal approximation. The sufficient statistic for $\boldsymbol{\beta}$ is $\mathbf{X}^\top \mathbf{y}$, by Lemma 7.4.1 applied to Eq. 35.3.1, so the conditional distribution of the component belonging to one coefficient, given the rest, depends on that coefficient alone. Enumerating it gives exact p -values and a median-unbiased estimate that is finite even under separation. The idea is Cox’s (1970) and the network algorithms are those of Hirji, Mehta and Patel (1987); the limit is computational.

Idea. Separation is not a defect of the data; it is the data saying that the effect is larger than the sample can measure, and every remedy decides how much larger. A one-sided likelihood ratio interval is the most honest of them, because it does not decide.

35.3.7 Exercises

35.3.7.1 A. Check your understanding

Exercise A1.

A model contains an indicator for one level of a categorical covariate, and every observation at that level has $y_i = 1$. Exhibit a separating direction, say which of the three cases of [Definition 35.3.1](#) holds, and describe the fit.

Solution. Let $\mathbf{b}$ have 1 in the coordinate of that indicator and zeros elsewhere. For observations at that level $s_i \mathbf{x}_{(i)}^\top \mathbf{b} = +1 > 0$; for the others $\mathbf{x}_{(i)}^\top \mathbf{b} = 0$. So $\mathbf{b}$ separates with some margins zero: quasi-complete separation, unless the remaining data are themselves completely separated. The coefficient of that indicator diverges, its fitted probabilities go to 1, and the other coefficients converge to the estimates from the remaining observations.

35.3.7.2 B. Practice

Exercise B1.

Show that for any data and any $\lambda > 0$ the function $\boldsymbol{\beta} \mapsto \ell(\boldsymbol{\beta}) - \frac{1}{2}\lambda\|\boldsymbol{\beta}\|^2$ has exactly one maximizer, without assuming $\text{rank}(\mathbf{X}) = p$. Show also that if $\text{rank}(\mathbf{X}) = p$ and both outcomes occur the same is true when the intercept is left unpenalized.

Solution. By [Eq. 35.3.1](#), $\ell \leq 0$, so the objective is at most $-\frac{1}{2}\lambda\|\boldsymbol{\beta}\|^2$, which tends to $-\infty$: the objective is coercive and continuous, so it attains its maximum on a compact sublevel set. It is the sum of the concave ℓ (Exercise (C1)) and a strictly concave quadratic, hence strictly concave, so the maximizer is unique. Neither step used the rank of $\mathbf{X}$. With the intercept β_0 unpenalized, write $\boldsymbol{\gamma}$ for the slopes. If $\|\boldsymbol{\gamma}\| \rightarrow \infty$ the penalty alone sends the objective to $-\infty$, since $\ell \leq 0$; if $\boldsymbol{\gamma}$ stays bounded and $|\beta_0| \rightarrow \infty$, then some observation has $s_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta} \rightarrow -\infty$, because an observation with $y_i = 1$ supplies one as $\beta_0 \rightarrow -\infty$ and one with $y_i = 0$ as $\beta_0 \rightarrow +\infty$. So the objective is again coercive, and strict concavity now comes from ℓ itself.

Exercise B2.

Take a 2×2 table with a single binary covariate, coded as an intercept and an indicator. Compute the h_i of [Proposition 35.3.2](#), and show that Firth's pseudo-data amount to adding $\frac{1}{2}$ to each of the four cell counts. What is the resulting estimate of the log odds ratio?

Exercise B3.

For a single binary covariate with fitted probabilities $\hat{\pi}_1$ and $\hat{\pi}_0$ in groups of sizes m_1 and m_0 , show that as $\hat{\pi}_1 \rightarrow 1$, with everything else fixed, the Wald statistic tends to zero while the likelihood ratio statistic increases to a positive limit.

35.3.7.3 C. Going deeper

Exercise C1.

Suppose the rows $\mathbf{x}_{(i)}$ are independent draws from a distribution symmetric about the origin in $\mathbb{R}^p$ and the responses are independent coin flips. Cover's counting argument gives the probability of complete separation as $2^{1-n} \sum_{k=0}^{p-1} \binom{n-1}{k}$. Deduce that separation is almost certain when $n < 2p$ and almost impossible when $n \gg 2p$, and relate this to [Chapter 28](#).

Solution. The sum is $\Pr\{B \leq p-1\}$ with $B \sim \text{Binomial}(n-1, \frac{1}{2})$, so with $E B = (n-1)/2$ it is near 1 when $p-1$ exceeds $(n-1)/2$, roughly $n < 2p$, and near 0 when p is well below $n/2$, the transition having width of order $\sqrt{n}$. So with pure noise maximum likelihood fails to exist below about $n = 2p$: twice the sample size at which least squares begins to interpolate in [Chapter 28](#). Both thresholds mark the point past which an unpenalized fit reproduces the labels exactly, and both are why penalization is the default in high dimensions.